

Experiment	<i>Lab DIY: Drag of Various Flap Angles of a 2D Wing</i>
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Summary

In this experiment, Embry-Riddle Aeronautical University's open-loop subsonic wind tunnel was used to perform a wake survey study on the effects of adjusting the flap angle on a 2D NACA 4412 airfoil wing. The flap was tested at five different angles while maintaining a constant angle of attack: 0° , 15° , 25° (half flap), 35° , and 50° (full flap). The velocity profile in the wake region behind the wing was measured using a transverse pitot-static probe, which enabled the drag force to be determined in each scenario. The ambient temperature (T_{amb}) was measured to be 294.15K, and the ambient pressure (P_{amb}) was measured to be 102268.94 Pa. The drag uncertainty varied from 0.0426 to 0.0587, which indicates low error. For the 0 degree flap deflection, the drag force was calculated to be 1.882N, for the 15-degree flap deflection, the drag force was calculated to be 1.964N, for the 25-degree flap deflection, the drag force was calculated to be 1.832N, for the 35-degree flap deflection, the drag force was calculated to be 1.928N, for the 50-degree flap deflection, the drag force was calculated to be 2.20N.

During the setup, some unexpected challenges occurred. Initially, it was difficult to mount the 2D wing onto the force balance due to a lack of proper screws, which led to delays as replacements needed to be searched for in the lab. Once the wing was mounted, the adhesive meant to hold it in place failed. This happened because the default angle of attack set on the balance was too high, and the balance couldn't resist the wing's own weight due to the angle, causing the wing to detach. First, there was an attempt to reattach it using super glue, but it didn't hold securely, so eventually it was mounted using electrical tape. Even with the tape, the wing was not secure and relied on the friction between the wingtip and the walls of the wind tunnel test section to help stabilize it. Additionally, one side of the wall was taped to provide more support. These makeshift fixes introduced some uncertainty into the setup, which was observed and taken into consideration during the analysis.

Despite these issues, the wake surveys were successful in capturing usable data. The results provided a clear comparison between the drag in flap-up and flap-down configurations, as well as for the intermediate flap angles. The data showed a consistent trend of increasing drag (induced) with greater flap deflection.

Aim & Objectives

To examine the effect of flap deflection on the drag experienced by a NACA 4412 airfoil 2D wing using wind tunnel wake survey techniques.

The Objectives are as follows:

- Measure velocity profiles in the wake of the airfoil at various flap deflections.

- Determine the drag coefficient and drag force that corresponds to each configuration.
- Examine the drag properties of the maximum (flap down) and minimum (flap up) deflections.
- Account for and reflect on setup difficulties and potential sources of error in interpreting results.

Uncertainty Analysis

Uncertainty plays a major role in this experiment because it allows for results to be projected in a range that provides 95% confidence. This must be done for all variables to ensure the results are reliable and consistent. The uncertainty of airspeed can be calculated using the propagation and sum of squares equation.

$$\delta U = \sqrt{\left(\frac{\partial U}{\partial P_{Dyn}} \delta P_{Dyn}\right)^2 + \left(\frac{\partial U}{\partial P_{atm}} \delta P_{atm}\right)^2 + \left(\frac{\partial U}{\partial T_{atm}} \delta T_{atm}\right)^2}$$

With the following being expressed as

$$\begin{aligned}\delta P_{Dyn} &= 1.96 \frac{S_{P_{Dyn}}}{\sqrt{N}} \\ \delta P_{atm} &= 16.9319 \text{ Pa} \\ \delta T_{atm} &= 0.2778 \text{ K}\end{aligned}$$

Furthermore, airspeed can be calculated using:

$$U = \sqrt{\frac{2P_{Dyn}}{\rho}} = \sqrt{\frac{2(P_{Dyn})(R)(T_{atm})}{P_{atm}}}$$

Taking partial derivatives used in uncertainty equations:

$$\frac{\delta U}{\delta P_{Dyn}} = \frac{1}{2} \sqrt{\frac{2(R)(T_{atm})}{(P_{atm})(P_{Dyn})}}$$

$$\frac{\delta U}{\delta P_{atm}} = \frac{1}{2} \sqrt{\frac{2(P_{Dyn})(R)(T_{atm})}{(P_{atm})^3}}$$

$$\frac{\delta U}{\delta T_{atm}} = \frac{1}{2} \sqrt{\frac{2(P_{Dyn})(R)}{(P_{atm})(T_{atm})}}$$

For the coefficient of drag the trapezoidal integration method in MATLAB was used, expressed as the following where U_∞ was the average freestream velocity and U_w was the velocity in the wake deficit.

$$\delta D'_{\rho_\infty} = (\rho_\infty + \delta \rho_\infty) \int_{h_1}^{h_2} U_w (U_\infty - U_w) dy$$

$$\delta D_{U_w}' = \rho_\infty \int_{h_1}^{h_2} (U_w + \delta U_w)(U_\infty - (U_w + \delta U_w)) dy$$

$$\delta D_{U_\infty}' = \rho_\infty \int_{h_1}^{h_2} U_w ((U_\infty + \delta U_\infty) - U_w) dy$$

$$\delta D' = \sqrt{(\delta D_{U_\infty}')^2 + (\delta D_{U_w}')^2 + (\delta D_{\rho_\infty}')^2}$$

All uncertainties were calculated using error propagation with 95% confidence. Errors due to precision or systematic errors were taken into consideration with MATLAB.

Experimental Setup

Flap Control

The experimental setup incorporated a flap control system consisting of a microcontroller interfaced with servos installed on the wing. Power to the servo system was regulated using an external DC power supply ensuring stable operation throughout the testing period. The controller allowed for **precise** adjustment of flap deflection angles in discrete increments.

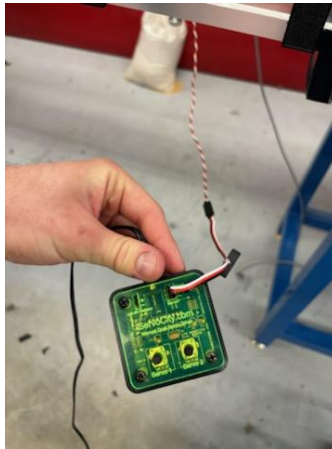


Figure 1. Servo Controller Setup

Test Section

The test section is a rectangular chamber with transparent walls to be able to see the instrumentation and/or test articles and is pictured in Figure 2. The wind tunnel is equipped with two different pitot tubes, one that is stationary and one that can be traversed around the test section. There is also a digital and analog manometer to provide pressure readings from inside the test section.

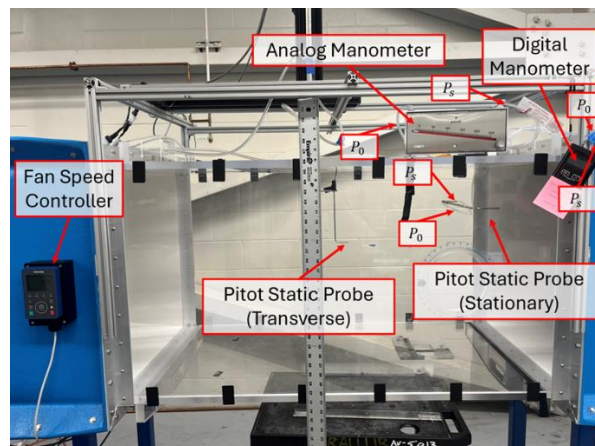


Figure 2. Test Section Setup

Data Acquisition System

The wind tunnel is integrated with a data acquisition managed by this computer station. The system includes the data acquisition computer (DAQ), the analog to digital converter (ADC), a pressure transducer, a transverse direction controller to control the location of the pitot probe, as well as stagnation and static pressure lines are all properly called out in Figure 3. There is a MATLAB script that is run on the DAQ to log and process the pressure data coming in, as well as automatically move the pitot probe in the test section.

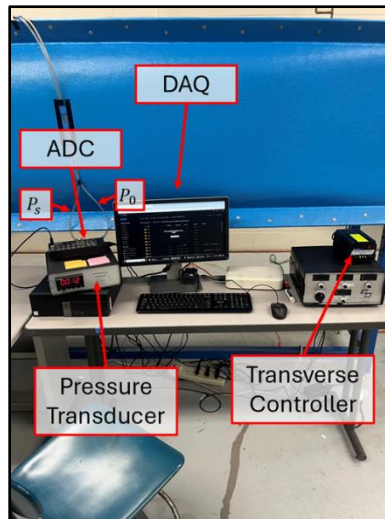


Figure 3. Computer Station Setup

Test Article Installed in Test Section

The model was mounted to the force balance and held at a constant angle of attack. With issues of structural stability previously discussed, electrical tape was used to further stabilize the test article to ensure the wing remained attached to the force balance.

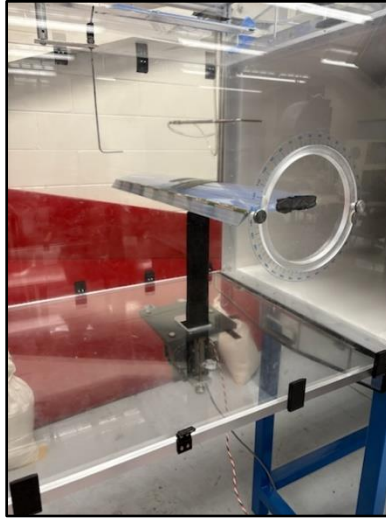


Figure 4. Full Flap Setup

Results & Discussion

The wake velocity profiles measured at varying flap deflection angles are presented in Figure 1. As flap deflection increased the wake structure downstream of the airfoil exhibited a notable change in velocity distribution. Lower deflection angles exhibit a narrow wake velocity profile and as the deflection angle increases the wake profile gets wider. This trend suggests a rise in aerodynamic drag with increasing flap angle which is consistent with expected flow behavior over a deflected control surface.

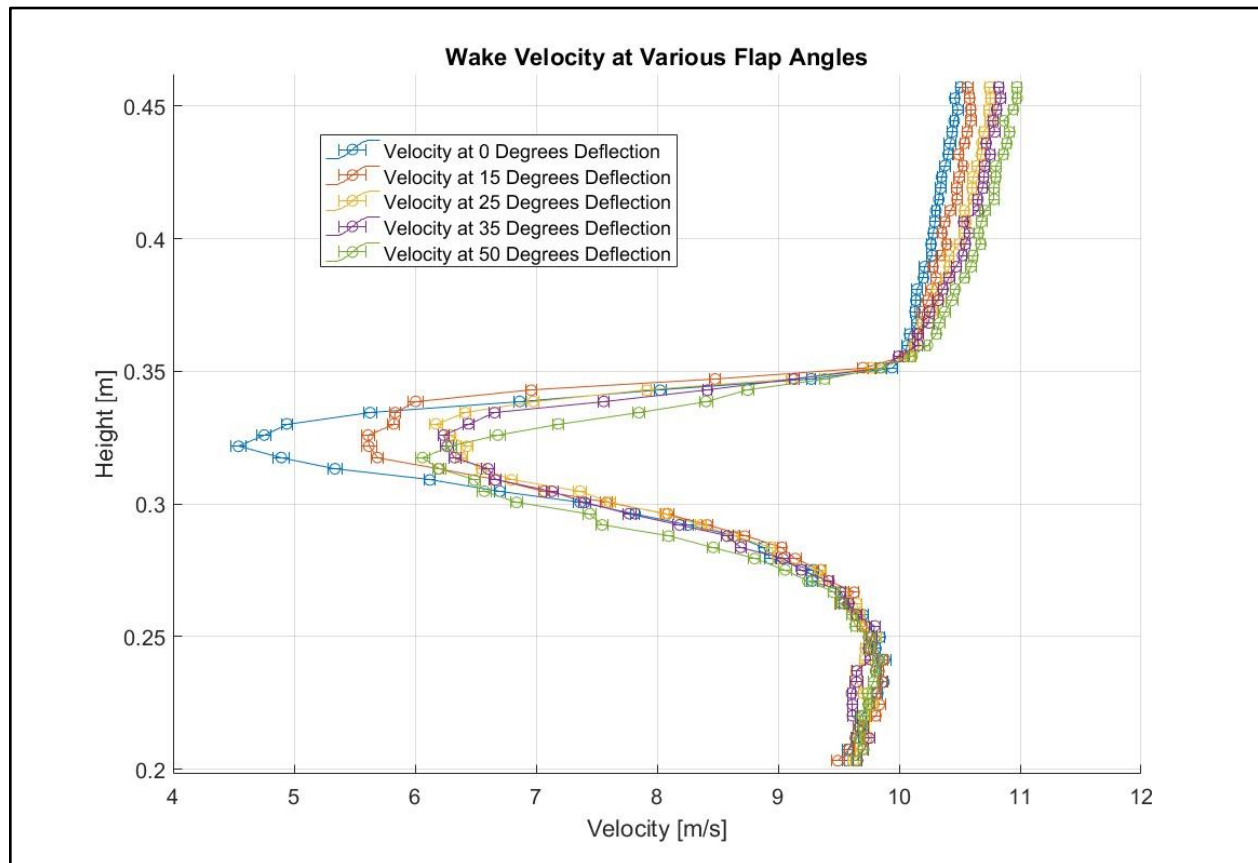


Figure 5. Wake Velocity at Various Flap Angles

Table 1 quantifies the corresponding drag forces measured at each flap angle along with their associated uncertainties. The drag generally increased with flap deflection rising from 1.8824 N at 0 degrees to a maximum of 2.200 N at 50 degrees; however, this trend was not strictly linear. At 25 degrees a slight dip in drag was observed to be 1.8328 N which could be attributed to experimental error or possibly minor flow reattachment. The uncertainties remained under 0.06 N for all trials, indicating reasonable repeatability of the measurements.

Table 1. Tabulated Results of Drag at Various Flap Deflection Angles

Flap Deflection [Degrees]	Drag [Newtons]	Uncertainty [Newtons]
0	1.8824	± 0.0443
15	1.9636	± 0.0493
25	1.8328	± 0.0426
35	1.9277	± 0.0482
50	2.2000	± 0.0587

To facilitate comparison across deflection angles the drag coefficient (C_d) calculated and plotted in Figure 2. The rise in C_d at larger deflection angles increased pressure drag due to the enhanced separation and wider wake regions visualized in Figure 1.

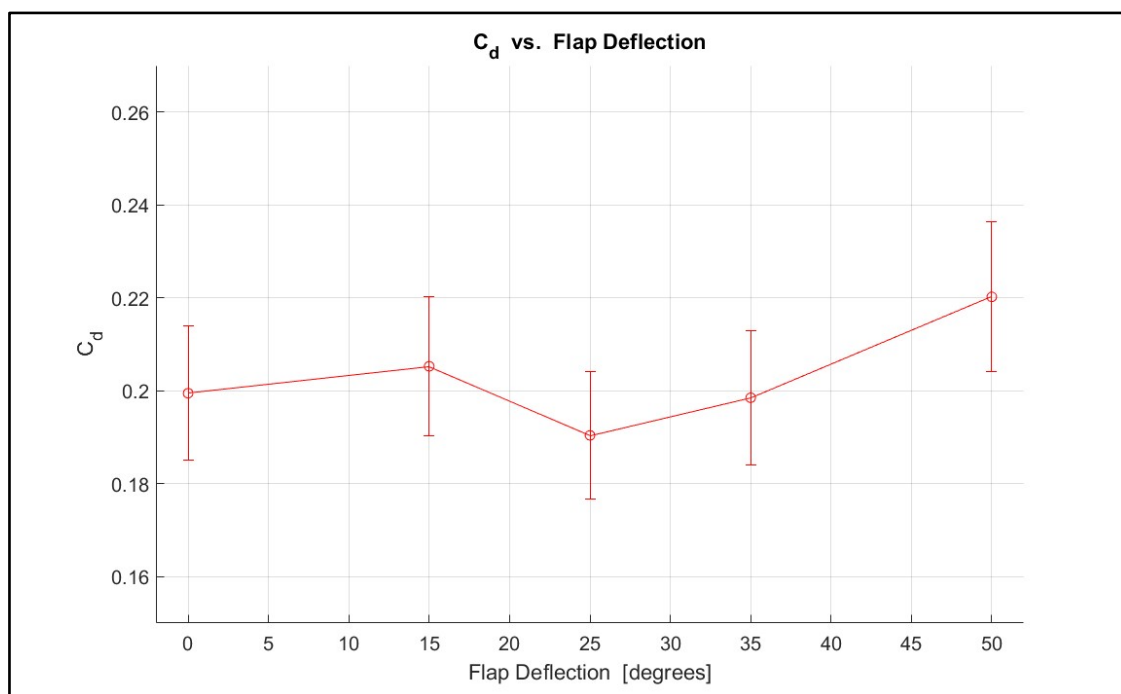


Figure 6. Coefficient of Drag at Various Flap Deflection Angles

Overall, the results align with theoretical expectations that increasing flap deflection increases drag due to enhanced flow disturbance and the wake profile becoming wider. The small irregularity at 25 degrees may warrant further investigation with finer angle increments or higher resolution wake measurements. Despite this, the results demonstrate a clear relationship between control surface deflection and aerodynamic performance emphasizing the tradeoff between lift and drag in flap deployment.

References

- [1] AE315 Supplement – Experimental Aerodynamics Supplement: (2025). Embry-Riddle Aeronautical University.

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Ambient

```
clear
clc
close all

P = 102269; %atm
T = 294.15; %k
R = 287;
rho = P/(R*T); %kg/m^3
c = 0.1524; %m
conversion = 248.84; %in of water to pa
flap = [0,15,25,35,50];
```

Loads wake

```
wake_av = zeros(61,5);

for i = 1:5
    if i == 1
        for j = 1:61
            wake_name = sprintf('Flap_angle_zero_10ms_%d.mat',j);
            wake_data = load(wake_name,'dp');
            wake_data = conversion*struct2array(wake_data);
            wake_av(j,i) = mean(wake_data);
            wake_std(j,i) = std(wake_data);
        end
    elseif i<5
        b = 15+10*(i-2);
        for j = 1:61
            wake_name = sprintf('Flap_angle_%d_10ms_%d.mat',b,j);
            wake_data = load(wake_name,'dp');
            wake_data = conversion*struct2array(wake_data);
            wake_av(j,i) = mean(wake_data);
            wake_std(j,i) = std(wake_data);
        end
    else
        for j = 1:61
            wake_name = sprintf('Flap_angle_0_10ms_%d.mat',j);
            wake_data = load(wake_name,'dp');
            wake_data = conversion*struct2array(wake_data);
            wake_av(j,i) = mean(wake_data);
            wake_std(j,i) = std(wake_data);
        end
    end
end
end
```

Filters wake

gets wake velocity

```
vwake = zeros(61,5);
for i=1:5
    for j = 1:61
        vwake(j,i) = sqrt(2*wake_av(j,i)/rho);
    end
end

% Loads wake spike
wake_spikes_data = struct2array(load("wakepoints.mat"));
wake_spikes_pos = vertcat(wake_spikes_data.Position);
y = (linspace(0,10,61)+8)*0.0254;

% Creates a velocity dataset without the wake spike
for i = 1:5
    a = 10 - (2*(i-1));
    b = 10 - (2*i-1);
    for j = 1:61
        if wake_spikes_pos(a,2) >= y(j) && wake_spikes_pos(b,2) <= y(j)
            meanready_v(j,i) = 0;
        else
            meanready_v(j,i) = vwake(j,i);
        end
    end
end
end
```

Drag

```
for i = 1:5
    u_inf(i) = mean(nonzeros(meanready_v(:,i)));
    q(i) = 0.5 * rho * u_inf(i).^2;
    yfilter = 0;
    uwfilter = 0;
    for j = 1:61
        if meanready_v(j,i) == 0
            wakedrag_v(j,i) = vwake(j,i);
            y_wd(j,i) = y(j);
        else
            wakedrag_v(j,i) = 0;
            y_wd(j,i) = 0;
        end
    end
    yfilter = y_wd(:,i);
    yfilter = yfilter(yfilter ~= 0);
    uwfilter = wakedrag_v(:,i);
    uwfilter = uwfilter(uwfilter ~= 0);
    D(i) = rho*trapz(yfilter,uwfilter.*(u_inf(i)-uwfilter));
    Cd(i) = D(i)/(0.5 * rho * (u_inf(i))^2 * c);
end
```

Uncertianity Propagation

```
dT = 0.5; %K
dP = 169.32; %Pa
drho = sqrt((1/(R*T)*dP)^2+(-P/(R*T^2)*dT)^2);
```

```

dc = 1/16*0.0254/c; %Non dimensional

for i = 1:5
    for k = 1:61
        dq_wake(k,i) = 1.96*wake_std(k,i)/sqrt(200); %pa
        dvwake(k,i) = sqrt((-sqrt(2*vwake(k,i))/2*(rho)^(-1.5)*drho)^2+(sqrt(1/(2*rho*vwake(k,i)))*dq_wake(k,i))^2);
        dq(i) = mean(dq_wake(:,i));
    end
end

for i = 1:5
    yfilter = 0;
    uwfilter = 0;
    for j = 1:61
        if meanready_v(j,i) == 0
            dvwake_drag(j,i) = dvwake(j,i);
            d_mrv(j,i) = 0;
        else
            dvwake_drag(j,i) = 0;
            d_mrv(j,i) = dvwake(j,i);
        end
    end
    yfilter = y_wd(:,i);
    yfilter = yfilter(yfilter ~= 0);
    uwfilter = wakedrag_v(:,i);
    uwfilter = uwfilter(uwfilter ~= 0);
    dvwake_filter = dvwake_drag(:,i);
    dvwake_filter = dvwake_filter(dvwake_filter ~= 0);
    du_inf(i) = mean(nonzeros(d_mrv(:,i)));
    dv(i) = mean(dvwake(:,i));
    dDrho(i) = (rho+drho)*trapz(yfilter,uwfilter.*(u_inf(i)-uwfilter)) - D(i);
    dDuw(i) = rho*trapz(yfilter,(uwfilter+dvwake_filter).*(u_inf(i)-(uwfilter+dvwake_filter))) - D(i);
    dDu_inf(i) = rho*trapz(yfilter,uwfilter.*(u_inf(i)+du_inf(i)-uwfilter)) - D(i);
    dD(i) = sqrt(dDu_inf(i)^2+dDuw(i)^2+dDrho(i)^2);
    dcd(i) = sqrt((dD(i)/q(i)/c)^2+(-D(i)/c*q(i)^-2*dq(i))^2+(-D(i)*c^-2/q(i)*dc)^2);
end

```

plots

```

figure(1)
for i = 1:5
    hold on
    fontsize(12,'points')
    errorbar(vwake(1:61,i),y,dvwake(1:61,i),'-o','horizontal')
    fontsize(12,'points')
    xlabel('Velocity [m/s]')
    ylabel('Height [m]')
    ylim([min(y)-0.005,max(y)+0.005])
    if i == 1
        legend_num = 0;
    elseif i < 5
        legend_num = 15+10*(i-2);
    else
        legend_num = 50;
    end
    legend_plot{i} = sprintf('Velocity at %d Degrees Deflection',legend_num);
    grid on
end
legend(legend_plot{1},legend_plot{2},legend_plot{3},legend_plot{4},legend_plot{5})
title('Wake Velocity at Various Flap Angles')

```

```
figure(2)
hold on
fontsize(12,'points')
errorbar(flap,Cd,dcd,'-or')
xlabel('Flap Deflection [degrees]')
ylabel('C_d')
xlim([-2,52])
ylim([0.15,0.27])
title(' C_d  vs.  Flap Deflection ')
grid on
```

